

Week 3: Interactive Plots

- **Key Concepts:**

- **Advanced Visualization:** Moving beyond standard Excel charts to visualize statistical distributions (Violin/Joyplots) and multi-dimensional relationships (3D plots).
- **Data Cleaning & Wrangling:** The transition from "perfect" textbook data to messy real-world data. Filtering data and handling missing values (`NaN`).

- **Coding Techniques:**

- **Interactive Plotting:** Using `plotly.graph_objects` to create Violin Plots, Joyplots (Ridge Plots), and interactive 3D Scatter Plots with rotation capabilities.

- **Packages Introduced:**

- `pandas` : Used heavily for data manipulation (reading CSVs, filtering dates, data cleaning, creating DataFrames).
- `plotly` (**Express & Graph Objects**): The "best-in-class" tool introduced for creating high-quality, interactive visualizations (zooming, hovering, 3D rotation) that surpass standard Excel charts.
- `marimo` : Continued use as the reactive notebook environment, specifically highlighting features like variable persistence, sandboxing, and exporting to HTML for sharing.